

H2-O deficiency promotes regulatory T cell differentiation and CD4 hyperactivity

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

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Reviewed preprint posted

October 23, 2023 (this version)

Posted to bioRxiv

August 16, 2023

Sent for peer review

July 19, 2023

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Abstract

Regulatory T cells (Treg) are crucial immune modulators, yet the exact mechanism of thymic Treg development remains controversial. Here, we present the first direct evidence for H2-O, a peptide editing molecular chaperon for MHC class II antigen processing, on selection of thymic Tregs. We demonstrate that lack of H2-O in the thymic medulla promotes thymic Treg development and leads to an increased peripheral Treg frequency. Single-cell RNA-sequencing (scRNA-seq) analysis of splenic CD4 T cells revealed an enrichment of both effector-like Tregs and activated CD4 T cells in the absence of H2-O. H2-O has been shown to increase the presentation of high-affinity class II epitopes, thus leading to proper negative selection in the medullary thymus. In spite of this notion, our data support two concepts; a) lack of expression of H2-O in thymic medulla creates an environment permissive to Treg development and, b) that loss of H2-O drives increased basal auto-stimulation of CD4 T cells. These findings can help in better understanding of autoimmunity and design of therapeutics for treatment of autoimmune diseases.

eLife assessment

This paper seeks to understand how the presentation of peptides by medullary thymic epithelial cells may be regulated by the MHCI peptide loading modulator, H2-O, and how this may affect the selection of regulatory T (Treg) cells. Further work is needed to ensure that the findings are robust: currently the analysis of data is **inadequate** and inconsistencies in the reported findings are not placed in context with results from other groups. The current version does not provide sufficient support for the claims regarding the effects on Treg cell selection.

Introduction

T cells are key players in humoral immune responses. Upon infection with a pathogen, CD4 T cells utilize their T cell receptor (TCR) to survey peptides presented by professional antigen presenting cells (APCs) on class II molecules (pMHCI) for recognition. Identification of a cognate pMHCI complex by the TCR leads to CD4 T cell activation and ultimately clearance of the foreign pathogen.

Faulty activation, however, can lead to deleterious inflammation causing possible autoimmune diseases and cancer development. Therefore, multiple regulatory processes exist to ensure proper T cell activation.

Regulation of T cell activation begins during thymic development where immature thymocytes are screened for self-reactivity. Broadly divided into positive and negative selection, this process ensures that CD4 T cells expressing high avidity self-reactive TCRs are either eliminated (1), or become CD4 regulatory T cells (Tregs) (2). Critical to the screening process for identifying auto-reactive T cells is the Class II antigen processing machinery expressed by medullary thymic epithelium cells (mTECs) and thymic antigen presenting cells (APCs). Two chaperone proteins, H2-M (murine; human, HLA-DM), and H2-O (murine; human, HLA-DO) are major components of the MHC II processing pathway. While H2-M is expressed in all APCs, H2-O is expressed mainly in thymic medulla and B cells (3). H2-M plays a critical role in MHC class II antigen processing by dissociating the Class II Invariant Chain peptide (CLIP) from the newly synthesized MHC II. Dissociation of CLIP promotes a peptide-receptive MHC II conformation that can readily scan for the best MHC II groove fitting epitopes of denatured antigens. However, a peptide-receptive MHC II conformation is highly transient (4, 5, 6, 7). We have proposed that H2-O binds to H2-M (8) and works cooperatively with H2-M by stabilizing the peptide-receptive MHC II conformation in order to optimize epitope selection (9). Together, these two molecules ensure that the best MHC II groove fitting epitopes are presented to cognate CD4 T cells.

While the exact mechanism of Treg selection remains to be fully understood (10), two critical requirements have been identified as necessary for successful thymic Treg development. First, thymic Treg development requires relatively strong TCR signaling in the thymic medulla (11), and second, Treg development relies on signaling by common γ chain (γ C) cytokines, mainly IL-2, for driving Foxp3 expression (12). Considering that strong TCR signaling during negative selection normally leads to CD4 T cell deletion (1), when investigating Treg selection, both the nature and density of medulla presented self-peptides must be considered. If epitopes are in high abundance and more ubiquitously expressed, then cognate CD4 T cells will undergo clonal deletion. However, if epitopes are in lower abundance and have a sparser expression, leading to discontinuous TCR stimulation, then cognate CD4 T cells might undergo Treg selection (10, 13, 14). This model of Treg selection relies on the level of TCR signaling that medulla localized CD4 single-positive (SP) T cells receive. Indeed, loss of H2-O function has been shown to alter both the repertoire and density of peptides presented by naïve peripheral B cells (15, 16). Thus, loss of H2-O could affect whether CD4 T cells are signaled for deletion, or survival during thymic deletion. Should a lower density of self pMHCII be presented by thymic epithelium, an increased number of self-reactive CD4 T cells might escape deletion leading to an increased frequency of auto-reactive T cell clones in the periphery. Or, alternatively, presentation of a lower density of self pMHCII could promote selection of regulatory T cells.

Here, we demonstrate that loss of H2-O generates a more stimulatory *in vivo* environment impacting both the thymic development and peripheral activation of regulatory T cells.

Results

Loss of H2-O increases the activation state of auditing medulla CD4 T cells

Previously, we demonstrated that loss of H2-O expression correlated with both an increased B cell presentation of low-affinity MHC II peptides, and an increased frequency of a MOG₃₅₋₅₅ specific, self-reactive CD4 T cell (15). Because of H2-O expression in the medullary thymus we speculated that H2-O deficiency might lead to presentation of lower densities of high-affinity self-peptides in the medulla, thereby causing an altered clonal deletion process. Medulla CD4 T cells can be

divided into two groups based upon expression of the pro-apoptotic protein Caspase-3 (17); “Auditing” CD4 T cells (CCR7+ CD4+ Caspase-3_{neg}) that are still undergoing self-auditing, and “Clonally Deleted” CD4 T cells (CCR7+ CD4+ Caspase-3_{pos}) that have been selected for thymic deletion.

We found that loss of H2-O significantly increased expression of the activation marker CD69 on auditing, but not CD4 T cells selected for clonal deletion (Fig 1a). CD69 in combination with MHC-I levels defines 3 medullary maturation stages: semi-mature (CD69+ MHC-I -), mature 1 (CD69+ MHC-I +), and mature 2 (CD69-MHC-I -) (17). Subdivision of auditing H2-O-KO CD4 T cells identified a significant increase in the Mature 1 stage (Fig 1b, middle). Conversely, both the semi-mature and mature 2 stages were decreased in H2-O KO mice (Fig 1b left/right). H2-O deficiency, however, did not appear to alter the rate of CD4 T cells undergoing clonal deletion (Fig 1c). Importantly, no differences were observed in thymocytes undergoing positive selection (sFig1). These data suggest that loss of H2-O drives a more stimulatory thymic medulla environment, but with similar levels of clonal deletion. It is therefore likely that the increased peripheral frequencies of MOG-specific CD4 T cells previously identified is due to increased peripheral expansion of the MOG-reactive clone, not a general altered clonal deletion.

H2-O KO thymi have increased regulatory T cell development

The increase in the Mature 1 (M1) state was noteworthy as Treg development has been shown to begin in the M1 stage (17). With increased levels of peripheral Treg previously identified in H2-O KO mice (15), we wondered if H2-O deficiency was affecting Treg selection. In fact, one model of thymic Treg selection centers around the concept of antigen density (13, 18). Within this “mosaic” model, sporadic MHC-TCR interactions with sparsely presented self-epitopes leads to Treg development. Having previously shown that peripheral loss of H2-O leads to increased presentation of lower affinity peptides (15), we postulated that altering the level of self-epitopes present in the medulla could alter Treg selection. In support of this idea, we found an increased frequency of CD25+Foxp3+ T cells in the H2-O KO CD4 SP population (Fig 2a). Furthermore, H2-O KO thymic Tregs (tTregs) expressed higher levels of CD25, CD5 and the orphan nuclear receptor Nur77 (Fig 2b). As Nur77 has been associated with the level of TCR engagement (19), increased Nur77 signal was particularly interesting since it indicates that absence of H2-O leads to increased TCR engagement suggesting an increased self-reactivity. We also found that H2-O KO tTregs were enriched in the M1 maturation stage (Fig 2c). This could mean two things; (1) loss of H2-O leads to increased presentation of specific Treg epitopes thereby increasing the frequency of Treg selection, or (2) lower antigen density allows an increased proportion of self-reactive CD4 T cells to engage their cognate ligands at suboptimal levels for clonal deletion but optimal ranges for Treg selection. Cumulatively, evaluation of H2-O KO thymi showed that only CD4 T cells undergoing the auditing process were affected by H2-O loss.

Loss of H2-O correlates with increased peripheral CD4 T cell activation

In light of the observation that lack of H2-O KO did not alter clonal deletion frequencies, we next evaluated whether peripheral loss of H2-O increased CD4 T cells activation. We found that H2-O KO spleens had a dramatic increase in the proportion of CD3+ CD4 T cells (Fig 3a). Critically, unimmunized H2-O KO spleens had increased levels of CD4 T cells co-expressing the key activation markers CD44+ and CD69+ (Fig 3b). We further assessed the levels of “non-activated” (CCR7+ CD62L+) versus “activated” (CCR7-CD62L-) CD4 T cells (20), and found an overall decreased percentage of CCR7 expressing CD4 T cells in H2-O KO spleens (Fig 3c) and lower frequencies of non-activated CD4 T cells in H2-O KO mice (Fig 3d). Importantly, this correlated with an

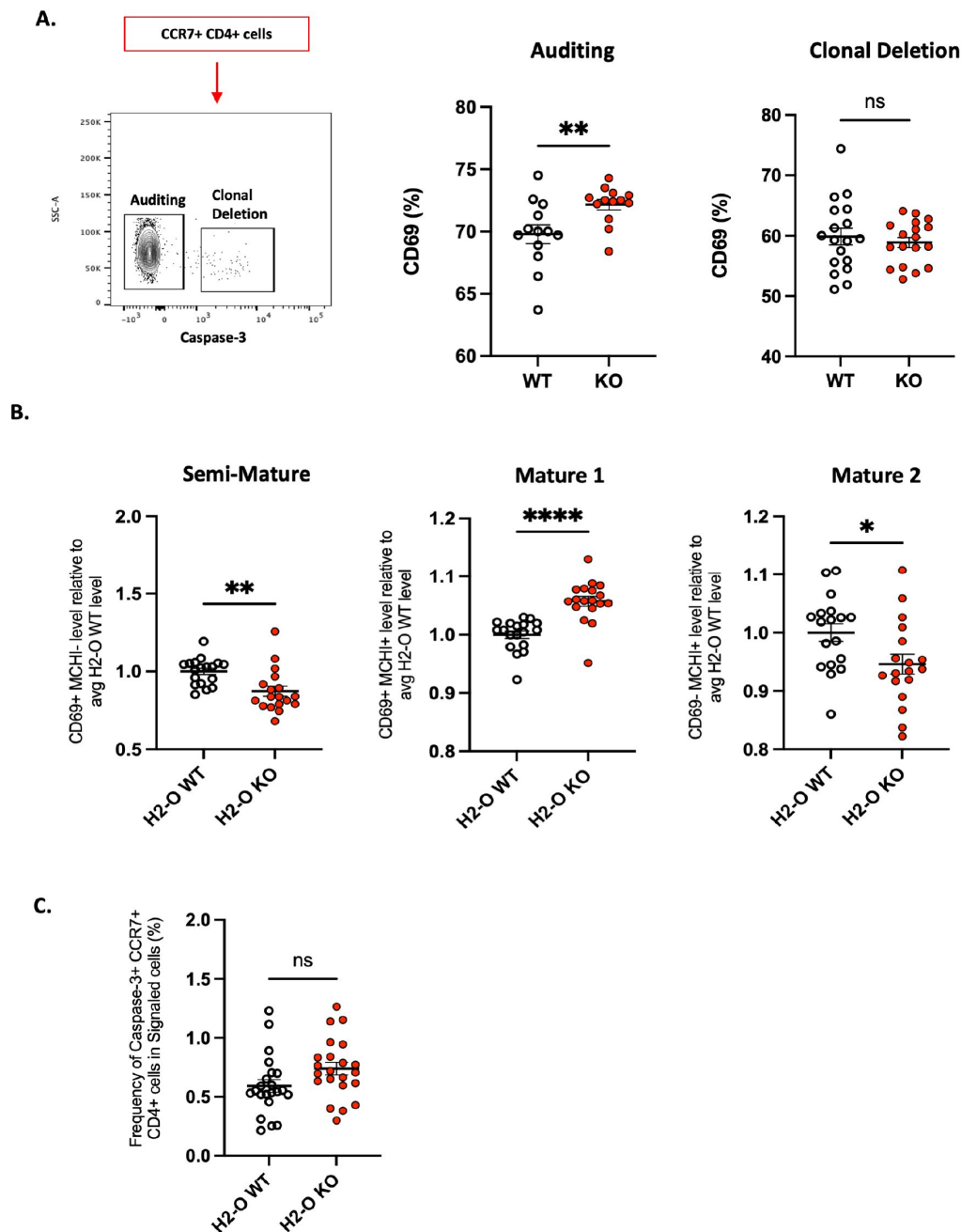


Figure 1

Loss of H2-O function supports increased thymic Regulatory T cell development

A. Left: representative gating showing how “Auditing” vs “Clonal Deleted” CCR7+CD4+ T cells are defined. Middle: Frequency of the activation marker CD69 in either “auditing” medulla CD4+ T cells or (Right) “clonally deleted” medulla CD4+ T cells.

B. Subdivision of auditing CD4 T cells into three maturation stages: Semi-Mature (Left), Mature 1 (Middle), and Mature 2 (Right). Expression has been normalized to the average H2-O WT levels within each experiment to allow for comparison across experiments. Data from 5 replicate experiments.

C. Frequency of medulla specific (CCR7+) CD4 T cells selected for clonal deleted (Caspase-3+) ns = not significant

* = <0.05

** = <0.001

*** = <0.0001

**** = <0.00001

Statistics: simple student T-test

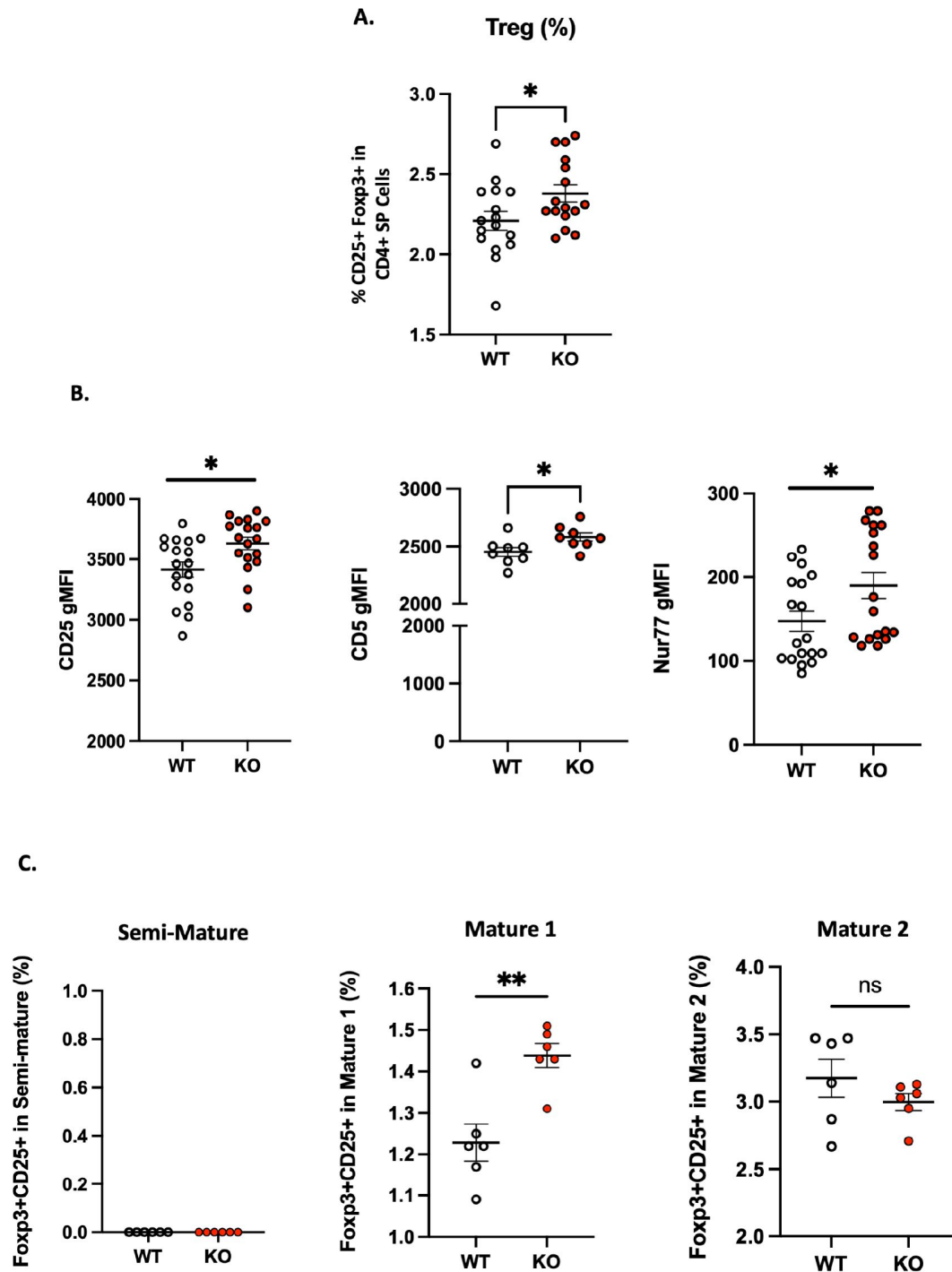


Figure 2

DO-KO thymi have an increased proportion of medulla CD4 T cells at the Mature 1 stage

A. The percentage of Foxp3⁺ CD25⁺ cells within the CD4 single-positive thymus population in H2-O WT (white) or H2-O KO (red) cells. Data from 3 replicate experiments

B. Geometric mean fluorescence intensity (gMFI) of CD25 (left), CD5 (right) and Nur77 (Bottom) expressed by Foxp3⁺ CD4⁺ T cells in the thymus of H2-O WT (white) or H2-O KO (red) mice.

C. Subdivision of Treg cells into three maturation stages: Semi-Mature (Left), Mature 1 (Middle), and Mature 2 (Right). Expression has been normalized to the average H2-O WT levels within each experiment to allow for comparison across experiments. Data from 2 replicate experiments.

increase in percentage of activated CD4 T cells (**Fig 3d**). Collectively, these phenotypic analyses support the idea that loss of H2-O leads to increased basal levels of activated CD4 T cells in unimmunized H2-O KO mice.

As discussed above, H2-O KO thymi promoted Tregs selection. Consistent with these observations and our previously published data (15, 21), H2-O KO spleens had an increased frequency of CD25⁺ Foxp3⁺ Tregs (**Fig 4a**). Furthermore, H2-O KO Tregs showed decreased levels of CD62L expression indicating these cells are likely more activated and circulating through the periphery (**Fig 4b**). Evaluation of the intracellular levels of Nur77 expression again revealed an increase, suggesting a higher TCR stimulation (**Figure 4c**) further validating the observations that loss of H2-O is driving a more activated state in both conventional CD4 T cells and Treg cells.

Single cell RNA-Sequencing of H2-O KO splenic CD4 T cells confirms increased activation

Based upon the strong FACS data above exhibiting increased numbers of Tregs, and more activated CD4 T cells, we attempted single-cell RNA-sequencing (scRNA-seq) to gain a more holistic unbiased characterization. CD3⁺ CD4⁺ NK1.1-CD19⁻ cells were sorted from spleens of 3 unimmunized H2-O WT and 3 H2-O KO mice and subjected to 10x Genomics scRNA-seq analyses. Principle Component Analysis (PCA) of the normalized scRNA-seq data identified 11 distinct CD4 T cell clusters (**Fig 5a** & **sFig 2**). Separation of the clusters based upon H2-O WT or H2-O KO genotypes identified a dramatic increase in cluster 0, cluster 2 and cluster 3 in the H2-O KO samples. Conversely, clusters 1 and 4 were significantly increased in H2-O WT samples (**Fig 5b**).

Comparing the genes from Clusters 0, 1 and 5 to the published CD4 T cell datasets using the “My Geneset” function on the Immunological Genome Project (www.imgen.org (<http://www.imgen.org/>)) suggested that upregulated genes ($\text{Log}_2\text{Avg Fold-change (FC)} \geq 0.5$) were present in both “naïve” and “activated” CD4 T cell datasets. Based upon the high expression of CCR7 however, these clusters were labeled as “non-activated”. Clusters 3, 6, 7, and 11 aligned with mainly an “activated” CD4 T phenotype. Cluster 3 was found to have increased expression of the activation marker CD44, while Cluster 11 expressed high Ki67. Clusters 2 and 4 contained the known Treg genes Foxp3 and Il2ra (CD25) (**Fig 5a** & **Sup Table 1**). Clusters 8, 9, and 10 were small populations of Macrophages and NKT cell (**Sup Table 1**). Condensing the cluster analysis further supported our initial FACS data that unimmunized H2-O-KO mice have a decreased frequency of “Non-activated” CD4 T cells (68.90% H2-O KO vs 74.17% H2-O WT with a coinciding increase in “Activated” (14.14% H2-O KO vs 8.53% H2-O WT) CD4 T cells (**Fig 5b**). To further test whether the higher activated state in H2-O KO T cells were due to the H2-O KO CD4 T cells themselves, or the environment, we performed an adoptive transfer experiment of dye labeled naïve (CD4⁺ CD44- CD25⁻) Thy1.1⁺ CD4 T cells from H2-O WT into either unimmunized Thy1.2⁺ H2-O WT, or Thy1.2⁺ H2-O KO hosts. As shown in **Fig 5d**, H2-O KO hosts induced significantly more *in vivo* proliferation of the WT cells than the H2-O WT hosts. These data additionally confirms that loss of H2-O creates a more activating CD4 T cell environment by presenting a wider range of self-epitopes (15).

H2-O KO Tregs have a more activated phenotype

Analyses of the two Treg clusters revealed that the frequency of Clusters 2 and 4 were roughly equal in H2-O WT samples, whereas H2-O KO samples had an overrepresentation of Cluster 2 (**Fig 6a**, **left**). Both cluster 2 and 4 expressed the classic Treg genes Foxp3, IL2ra [CD25], CTLA4, and Pdcd1 [PD-1] (**Fig 6a**, **right**; **Sup Table 1**). Also identified was the transcription factor Ikzf2 [Helios], a marker of stable Treg lineage commitment in inflammatory conditions (22) and debated marker of thymic Tregs (23, 24, 25, 26). Supporting our scRNA-Seq analysis, Helios protein expression was found in both H2-O WT and H2-O KO CD4 Tregs (**Fig 6b**). The data also showed significant Helios expression in H2-O KO Foxp3-CD4 T cells supporting that loss of H2-

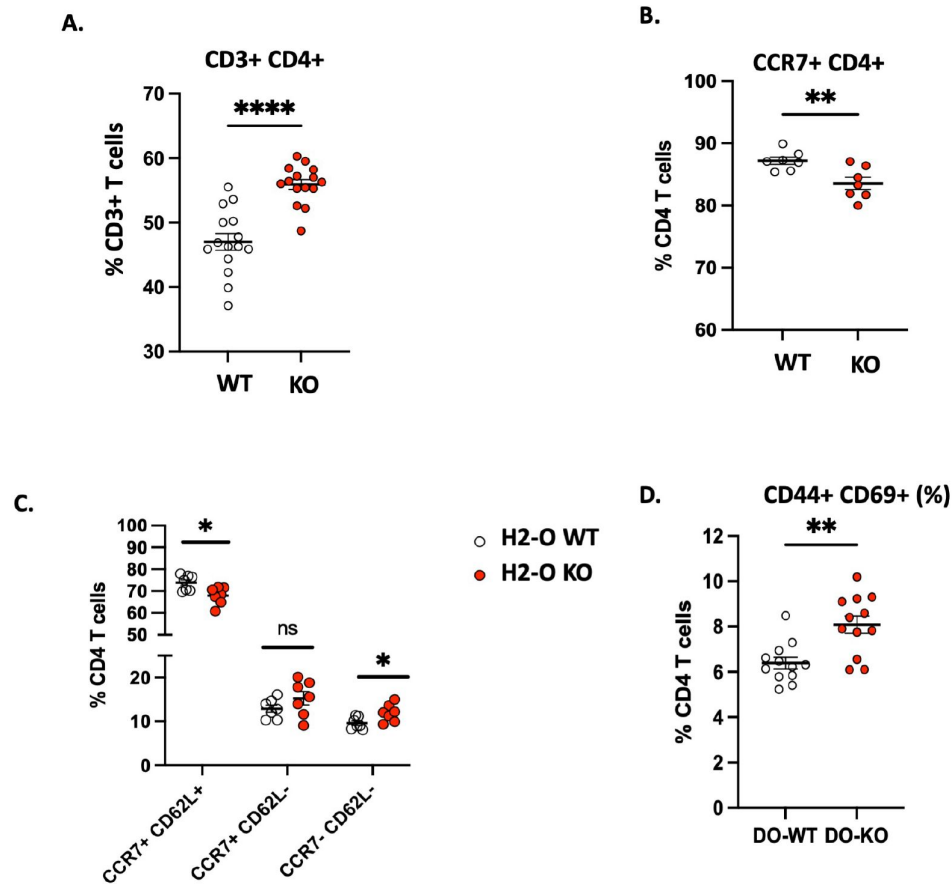


Figure 3

H2-O KO mice have an increased activated peripheral CD4 T cell population

A. Percentage of CD4+ T cells within the splenic CD3+ population of unimmunized 6-10-week-old H2-O WT (white) and H2-O KO (red) mice. Representative of >4 independent experiments, N = 16 mice per group.

B. Percentage of CD4+ T cells expressing the lymphoid tissue homing receptor CCR7

C. CCR7 and CD62L expression levels in unimmunized splenic CD4 T cells

D. Unimmunized levels of CD44 and CD69 expressed by splenic CD4 T cells. Increased CD69 expression, a marker of recent activation showed increased levels on H2-O KO CD4 T cells

* = <0.05

** = <0.001

*** = <0.0001

**** = <0.00001

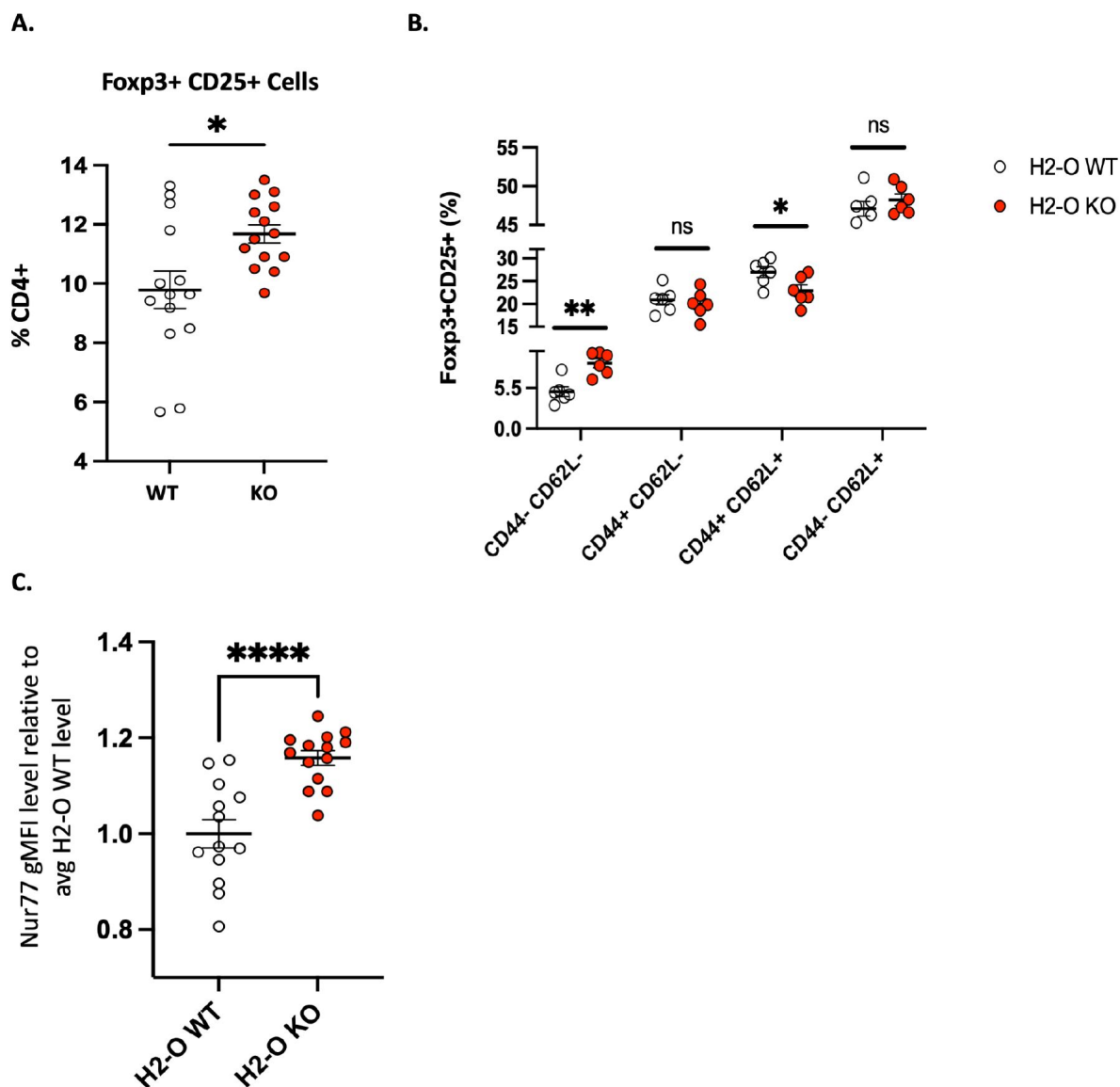


Figure 4

Peripheral H2-O KO Tregs have increased TCR stimulation

A. Frequency of Foxp3+CD25+ cells in splenic CD4+ T cell population of unimmunized mice
B. H2-O KO peripheral Tregs express decreased levels of naïve (CD44+ CD62L+) expressing cells
C. Normalized Nur77 gMFI levels in Foxp3+CD25+ Treg cells in H2-O WT (white) and H2-O KO (red) cells. To account for experimental variation the average Nur77 gMFI level in H2-O WT samples was calculated. gMFI levels in both H2-O WT and H2-O KO samples were then divided by the calculated H2-O WT average. An increased Nur77 ratio indicates increased Nur77 gMFI levels. Summary of 3 repeat experiments.

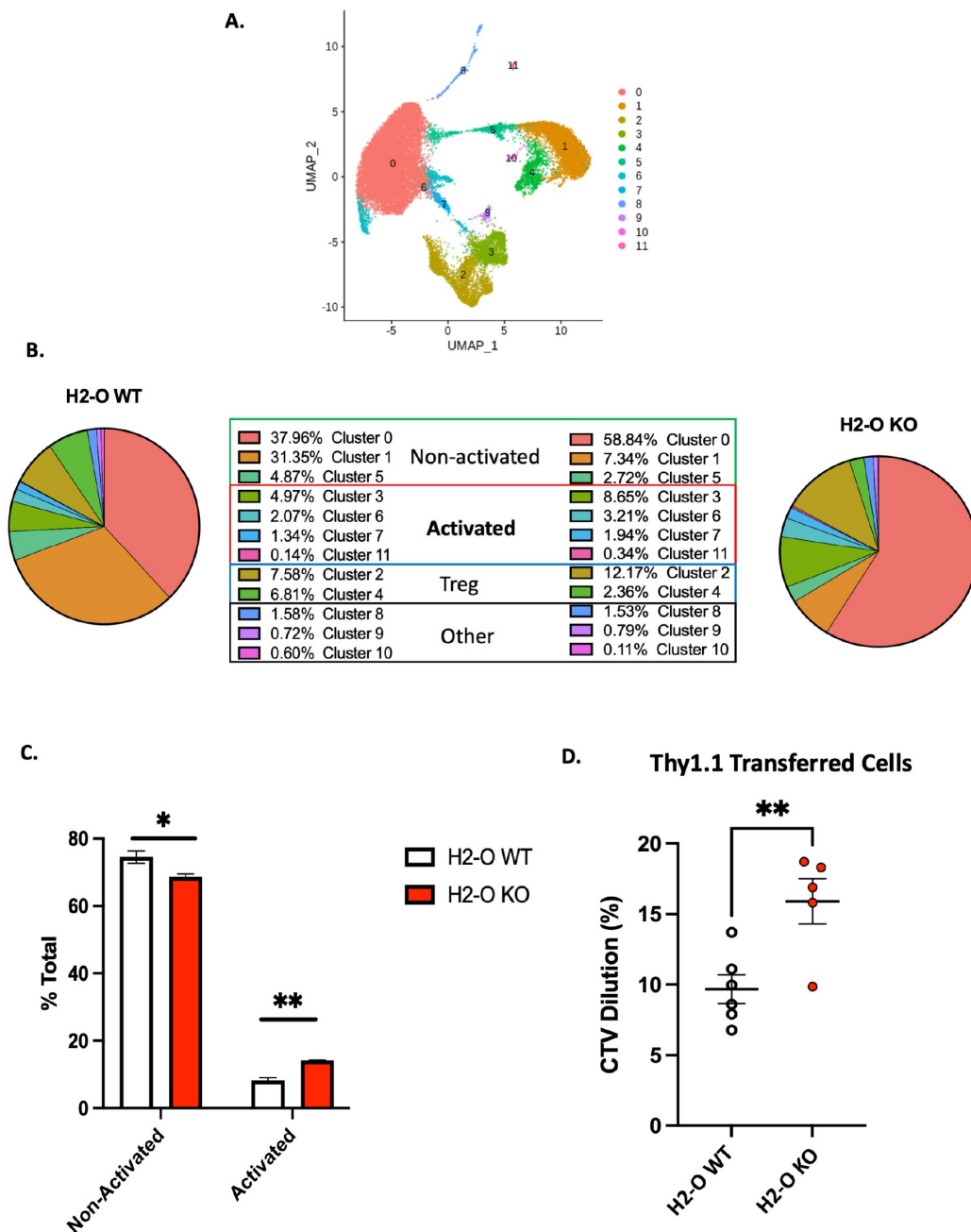


Figure 5

Loss of H2-O function causes increased basal CD4 cell activation

A. scRNA-seq clustering of CD4 T cells after Seurat analysis. Data represents the average of 3 biological replicates per genotype.

B. Breakdown of clusters in H2-O WT (Left) or H2-O KO (Right) samples. Clusters are grouped based upon, (1) known CD4 T cell subset markers and (2) gene comparison to published CD4 T cell data sets available on the Immunological Genome Project (www.immgen.org) (<http://www.immgen.org/>). Identified CD4 Cell phenotypes were: Non-activated, Activated, and Regulatory T cells. "Other" refers to a minor macrophage and NKT cell contamination from the sorting process.

C. Distribution of the for the Non-activated and Activated phenotypes across the H2-O WT and H2-O KO biological replicates (N = 3 mice per genotype).

D. In vivo proliferation of adoptively transferred naïve Thy1.1+ CD4 T cells (Thy1.1+ CD3+ CD4+ CD44-CD25-) after 7 days in either Thy1.2 H2-O WT (white) or Thy1.2 H2-O KO (red) hosts. Pooled data from 2 independent experiments (N= 6 H2-O WT, 5 H2-O KO)

O increases CD4 T cell activation. Interestingly, comparison of the cluster 2 genetic signature with a published Treg data set indicated that these cells are likely effector-like Tregs. Furthermore, breaking down Cluster 2 by genotype showed that H2-O KO Tregs had a larger fold change in expression of the core effector genes signature (27 [↗](#)) (**Fig 6c** [↗](#)). We hypothesize that this activated Treg state is likely driven by the increased basal CD4 T cell activation. Expansion of this Treg population could certainly be a mechanism controlling spontaneous autoimmunity in H2-O KO mice.

In summary, both scRNA-seq and FACs analyses showed that H2-O KO mice have an increased frequency of both conventional and regulatory CD4 T cells in the spleens of unimmunized mice. Loss of thymic H2-O expression correlated with an increased tTreg population that have received increased TCR stimulation which suggests increased autoreactivity. Furthermore, increased activation of conventional splenic CD4 T cells is likely a driving factor for the increased effector-like Treg status that was identified by scRNA-seq analyses of H2-O-KO CD4 cells.

Discussion

In this study, we have refined our understanding of how loss of H2-O alters CD4 T cell activity *in vivo*. Thymic analysis of medulla localized CD4 T cells pointed to a novel role for H2-O on CD4 T cells undergoing self-auditing. Importantly, lack of H2-O expression led to an increased frequency of both thymic and peripheral Tregs.

Tregs play a fundamental role in maintenance of homeostasis, hence a better understanding of the mechanism of their development is highly desirable and the subject of numerous studies (13 [↗](#), 18 [↗](#)). Our studies here are the first attempt in demonstrating that H2-O is a critical player in Treg development. While the exact mechanism by which thymic Treg cells are selected remains somewhat uncertain, recent studies support the idea that intermittent TCR signaling along with cytokine signaling drive Treg development (14 [↗](#)). These findings and others (28 [↗](#), 29 [↗](#), 30 [↗](#), 31 [↗](#), 32 [↗](#)) support a more avidity based model of Treg selection, in which alterations to the density of self-ligands present in the medulla will have differential effects on CD4 T cell selection. Intermittent TCR signaling as the driving force in Treg selection is very relevant to our understanding of how HLA-DO contributes to epitope selection during antigen processing. HLA-DO in complex with HLA-DM leads to better refinement of the epitopes from denatured antigens, promoting selection of the best fitting epitopes in the groove of MHC II molecules (9 [↗](#), 33 [↗](#)). As such, when DO is present, higher affinity peptides are more likely to be presented. Conversely, in the absence of DO a larger percent of lower affinity epitopes are selected. Indeed, we have recently reported that peptides eluted from H2-O KO mice expressing either murine I-A^b (15 [↗](#)) or human HLA-DR1 (Welsh et al, unpublished data) were of a lower general affinity. Accordingly, presentation of a larger portion of lower affinity peptides in the thymic medulla of H2-O KO mice is expected. Intermittent TCR signaling is typically generated by either pMHC II that are less stable (34 [↗](#), 35 [↗](#), 36 [↗](#), 37 [↗](#)), or TCR/pMHC of lower affinity (38 [↗](#)). Our findings here support the former. We associate increased Treg development as a consequence of the presentation of lower affinity peptides to self-auditing CD4 T cells.

Due to the continuous egress of mature CD4 T cells from the thymus, we also analyzed splenic CD4 T cells from unimmunized H2-O WT and H2-O KO mice. H2-O KO splenic CD4 T cells in unimmunized mice revealed an increased frequency of CD44^{hi}CD69⁺ antigen experienced CD4 T cells, which suggests an increased level of antigen-specific TCR signaling in H2-O KO mice. As previously implied (39 [↗](#)), a simple explanation for this activated state is incomplete thymic deletion in H2-O KO mice. While we were able to show that H2-O KO mice failed to delete specific CD4 clonal populations (15 [↗](#)), detection of differences at the global levels did not show significant changes between the two genotypes. Nonetheless, we suggest that presentations of different arrays of self-antigens as well as their lower densities on thymic medulla leads to a less than optimal

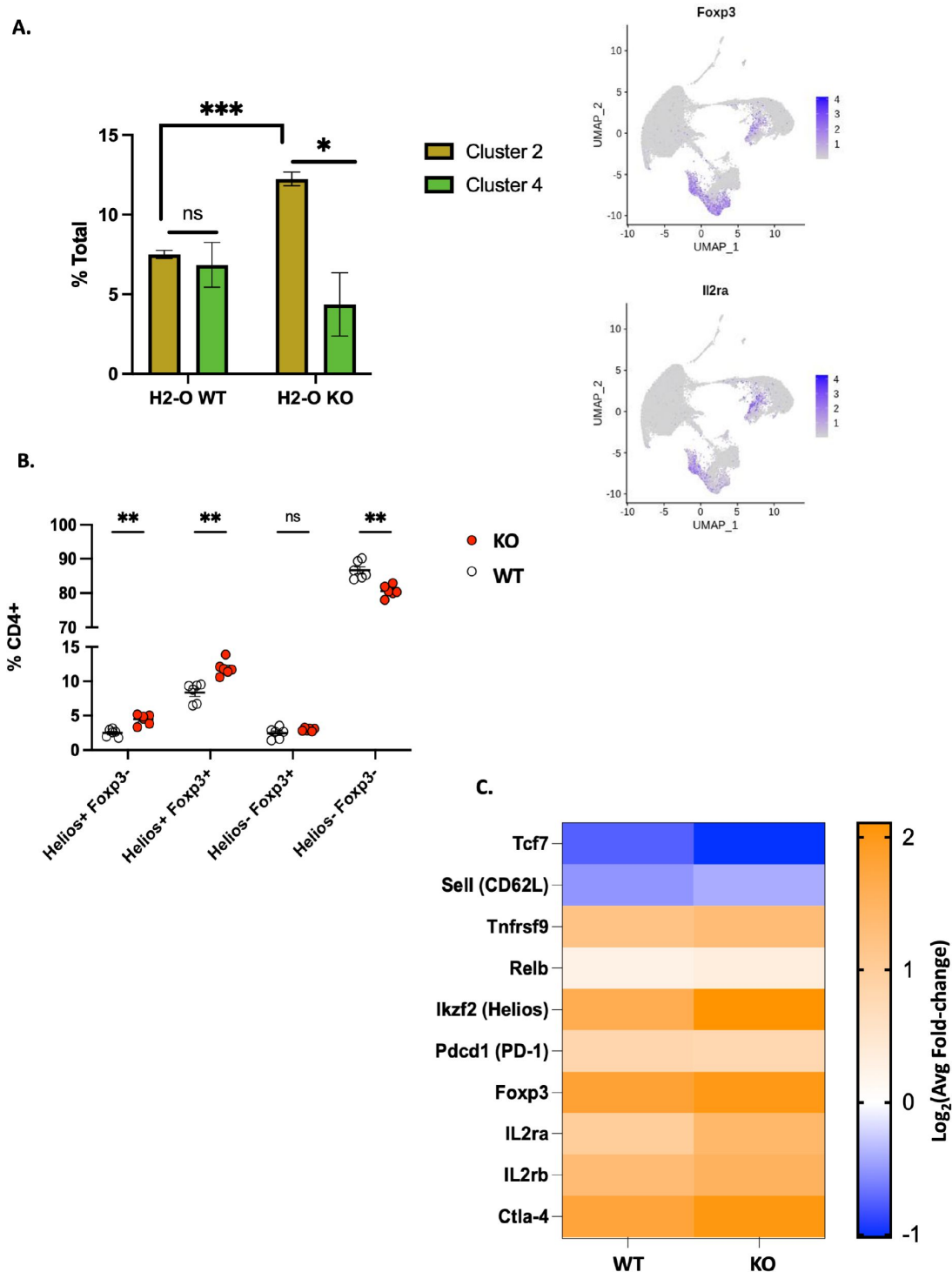


Figure 6

The H2-O KO Treg population has a more effector like phenotype

A. (Left) distribution of the two Treg populations identified by scRNA-seq clustering in either H2-O WT (left) or H2-O KO (right) samples. (Right) Expression of key Treg phenotypic markers: Foxp3 (Top) and CD25 [Il2ra] (Bottom)

B. Subdivision of the splenic CD4⁺ T cell population by Foxp3 and Helios expression

C. Comparison of cluster 2 upregulated genes ($\text{Log}_2\text{Avg FC} > 0.5$) in H2-O WT and H2-O KO samples to a published (Miragaia *et al.*) splenic effector Treg genetic profile

thymic deletion of self-reactive CD4 T cells and their routing to the periphery. Similar to activated CD4 T cells in periphery, H2-O KO Treg cells also had a more effector-like phenotype, indicating enhanced Treg activation in unimmunized mice (27 [↗](#), 40 [↗](#)).

In conclusion, our studies add a new dimension to our understanding of the role of H2-O in both CD4 T cell selection and activation. For the first time, we report on thymic negative selection in H2-O KO mice and demonstrate that loss of H2-O enhances thymic selection of regulatory T cells. Once in the periphery, an increased proportion of H2-O KO Tregs appear to be activated in a TCR dependent manner. These effector-like Tregs will likely help control increased basal CD4 T cell activation. However, the exact mechanism by which increased Treg cells are selected in the H2-O KO thymus remains to be determined. While we propose that alterations in medulla pMHCII-TCR avidity interactions could lead to enhanced Treg selection, increased presentation of self-antigens in a more tissue restricted antigenic (TRA) manner could also be possible (13 [↗](#)).

Methods

Mice

Male and female C57BL/6J (H2-O WT, stock no: 000664), Female B6.PL-Thy1a/CyJ (stock no: 000406) were purchased from Jackson Laboratories and bred in house. Generation of the H2-O knock-out mice has been previously described (41 [↗](#)) and mice bred in house. All mouse procedures were approved by the Johns Hopkins University Animal Care and Use Committee and were following relevant ethical regulations.

Antibodies/reagents

Flow Cytometry

anti-CD3e (17A2), anti-CD4 (RM4-5), anti-CD5 (53-7.3), anti-CD8 α (53-6.7), CD11c (N418), anti-CD19 (ID3), anti-CD25 (PC61), anti-CD44 (IM7), anti-CD45R/B220 (RA3-6B2), anti-CD62L (MEL-14), anti-CD69 (H1.2F3), anti-CD197/CCR7 (4B12), anti-NK1.1 (PK136), anti-TCR β (H57-597), anti-TCR $\gamma\delta$ (GL3), anti-Cleaved Caspase 3 (D3E9) Cell signaling (Danvers, MA); Foxp3 (150D), anti-Helios (22F6), anti-Nur77 (12.14), Fixable viability dye eFluor 780 (eBioscience).

FACs Sorting

anti-CD3e, anti-CD4, anti-CD19, anti-NK1.1, Propidium iodide. Briefly, 30,000 live CD4 T cells [Gating: CD3⁺, CD4⁺, CD19⁻, NK1.1⁻, PI⁻] were FACs sorted from the spleens of 3 unimmunized 10 week-old H2-O WT and 3 unimmunized 10 week-old unimmunized H2-O KO female mice.

Cell staining

For cleaved caspase 3 staining (17 [↗](#)) homogenized mice thymocyte cells were stained with anti-CCR7/CD197 at a final dilution of 1:50 for 30 min at 37°C prior to additional surface stains. Following surface staining, cells were fixed with Cytofix/Cytoperm (BD Biosciences) for 20 min at 4°C. Cells were then washed with Perm/Wash buffer (BD Biosciences) twice. Cells were stained with anti-cleaved caspase 3 at a 1:50 dilution at 23°C for 30 min.

For transcription factor staining, cells were incubated with surface antibody at 4°C for 20 min, permeabilized at 4°C for 30 min using a Foxp3/Transcription factor buffer set (Invitrogen, ThermoFisher Scientific), and then stained with anti-Foxp3 and/or anti-Helios at 23°C for 30 min.

Adoptive Cell Transfer

Naïve CD4 T cells were isolated from the pooled spleens and LNs of 4 week old Thy1.1 expressing C57BL/6J mice using the EasySep Mouse Naive CD4 T cell Isolation Kit (StemCell Technologies). Isolated cells were stained with eFluor 450 viability dye (eBiosciences) according to manufactures directions. 3×10^6 dye labeled Thy1.1 cells were IV injected into 10 week old H2-O WT or H2-O KO hosts. Transferred cells were recovered 7 days post-transfer.

scRNA-Sequencing

Library & sequencing

The samples were prepared using the 10x Genomics Chromium Next GEM Single Cell 5' Library and Gel Bead Kit v1.1, Chromium Next GEM Chip and Dual Index Kit TN Set A. They were run on the Illumina NovaSeq6000 with a run configuration of 28bp x 10bp x 10bp x 91bp.

Analysis

Analyses of T-cell scRNA-seq were performed with the package Seurat ([42](#)), as follows. Data was filtered to remove cells with low gene count (<200), large number of UMIs (>12,000) and high (>5%) fraction of mitochondrial reads. Expression levels of genes were log-normalized, and the most variable 2000 genes were selected for linear dimensionality reduction with Principal Component Analysis (PCA). The first 15 principal components were then used to performed unsupervised clustering using the Seurat SNN clustering package, with a resolution of 0.2. To identify cell types, potential markers for each cluster were calculated as the set of genes significantly differentially expressed in each cluster compared to all others, using the function FindAllMarkers, and by searching existing literature and marker databases. Lastly, differentially expressed genes for each cell type between conditions were determined using the function FindMarkers with the default function (bimod).

Defining clusters – Genes with an average fold-change (avgFC) >0.5 and an adjusted p-value <0.05 were run against CD4 T cell data sets available on the Immunological Genome Project (<https://www.immgen.org/>) using the “My Geneset” data browser function.

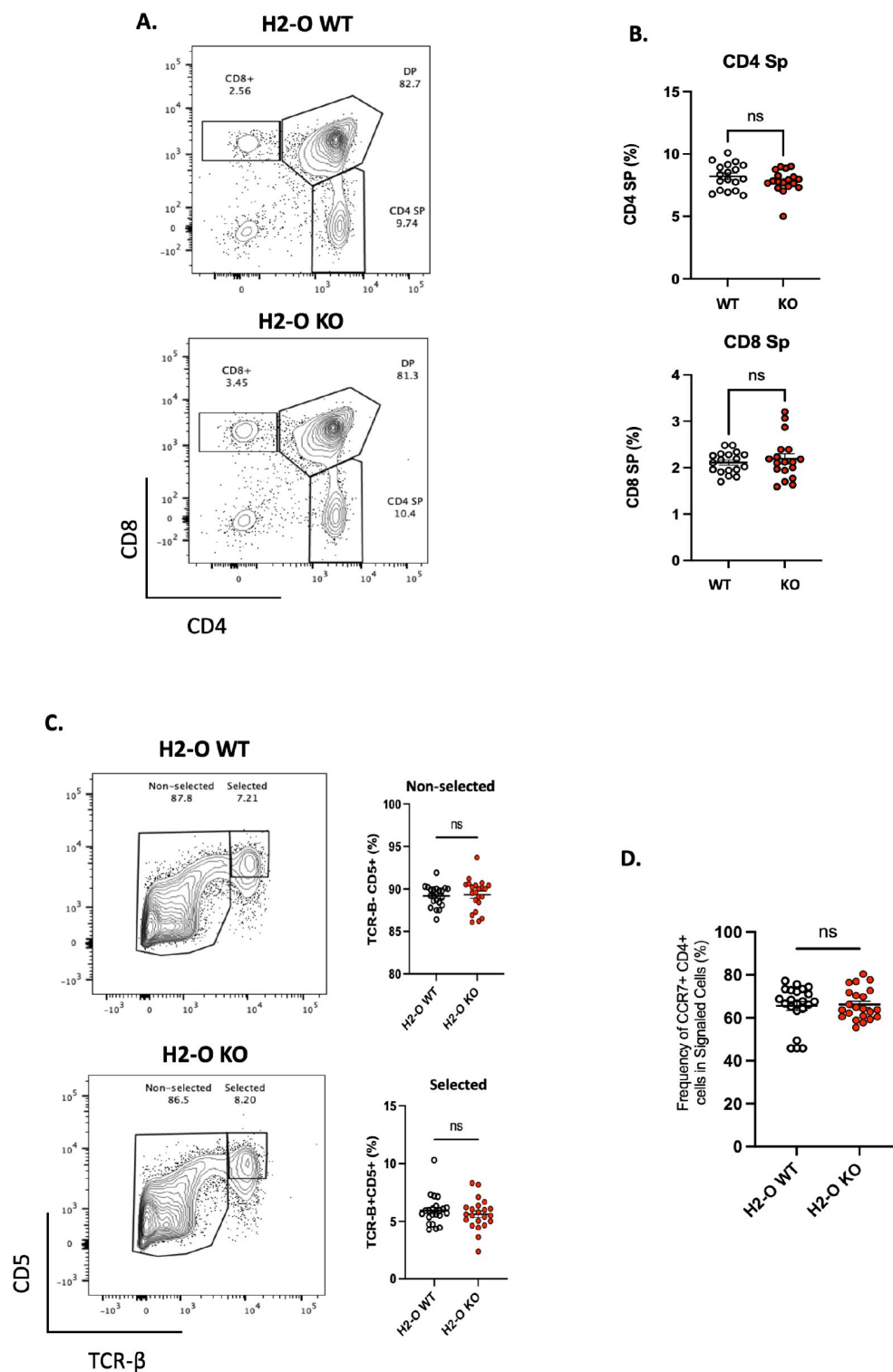
Statistical testing

GraphPad Prism was used for all statistical analyses. A standard Student T-test was used for estimation of statistical significance. Data is shown as mean $\pm$ SEM. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

Acknowledgements

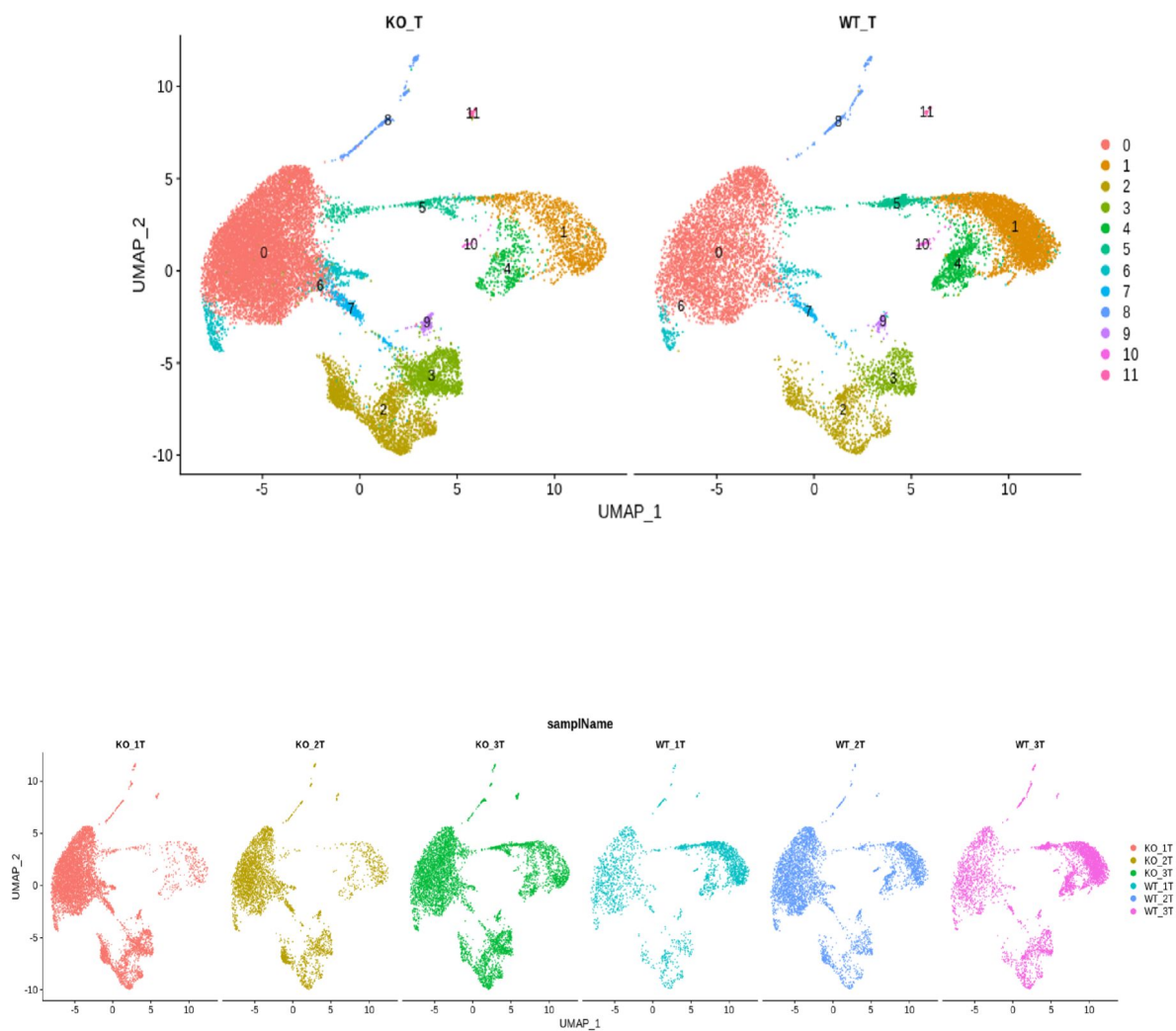
Supported by grants from NIAID, R01AI063764, R21AI101987, and R01AI120634, to SS-N. FACs analysis was performed in the BD Immunology and Flow Cytometry Laboratory at the Johns Hopkins School of Public Health.

Figure Legends



Supplemental Figure 1.

- A. Representative plots showing CD4, CD8 and Double Positive (DP) percentages in H2-O WT (Top) and H2-O KO (Bottom)
- B. Summary plots of the single-positive CD4 (Top) and CD8 (Bottom) percentages from 5 repeat experiments.
- C. (Left) representative flow plots showing the "Signaled" vs "Non-signaled" thymocytes in H2-O WT (Top) or H2-O KO (Bottom).
- (Right) Summary plots of >5 repeat experiments.
- D. Frequency of signaled CCR7+ medulla CD4 T cells in H2-O WT (white) and H2-O KO (red) mice



Supplemental Figure 2

Top: breakdown of H2-O KO (right) and H2-O WT (Left) Seurat clustering

Bottom: Seurat clustering for each individual biological replicate

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Reviewer #1 (Public Review):

The non-classical MHCII-like protein H2-M is essential for the loading of peptides on MHCII. The discovery that DM was partnered with a second MHCII-like protein, H2-O, which squelched or modified its activity was confounding. It was immediately speculated that H2-O was likely diminished self-peptide presentation. This led to the hypothesis that H2-O was involved in preventing unwanted CD4 T cell activation, thereby making autoimmunity less likely. 25 years of analysis of H2-O deficient mice have, indeed, shown that the self-peptide repertoire in the absence of H2-O is modestly altered. Demonstrating that autoimmunity results from this altered peptide repertoire has been decidedly less convincing. Old mice are reported to have increased serum anti-nuclear antibody titers, but mice prone to type 1 diabetes (T1D) and systemic lupus erythematosus (SLE) were not impacted by the loss of H2-O (Lee et al, 2021). Induction of the multiple sclerosis-like disease, EAE, in mice, was also shown to not be impacted by Lee et al 2021, although in a previous paper (Welsh et al 2020), the authors of this current manuscript suggest otherwise. Unfortunately, these discrepancies are not acknowledged by the authors, and the papers are, for the most part, not referenced.

In addition to antigen-presenting cells, H2-O is also found in MHCII-expressing medullary epithelial cells, suggesting it might play a role in T-cell selection. Direct data to support this

idea, however, has, at most, shown a minimal impact. In this manuscript, the authors follow up on their previous paper (Welsh et al, 2020) to further evaluate changes to T cell development. The conclusions are that H2-O impacts Treg development and changes the frequency and homeostasis of CD4 T cells. Although these would be interesting results, the data analysis is flawed, the presentation is incomplete, and the conclusions are exaggerated.

T-cell development analysis shown in Figs. 1 and 2 use the discovery from the Hogquist lab (Breed et al 2019) that thymocytes destined for clonal deletion can be differentiated from those still "auditioning" for selection by FACS for expression of cleaved caspase 3. Detection relies on complex FACS analysis that requires the exclusion of multiple populations, followed by accurate gating on CD5+TCRb+ cells (see Hogquist Fig. 1A). The authors apparently neglected to use the essential gating steps, but rather only used CD4 and CCR7 expression (Fig. 1A). This deviation from the Hogquist approach makes interpretation of Figs 1 and 2 meaningless. Even if this is an oversight in the description of the experiments, key conclusions are drawn from minimal changes to CD69 expression. CD69 is expressed as a continuum in the thymus (a "shoulder") making gating somewhat subjective and prone to variation from experiment to experiment. At the minimum, FACS data should be shown to indicate how these changes were measured, plus variations from mouse to mouse should be plotted, with statistics. FACS data needs to be shown to define how the complex semi-mature, M1, and M2 populations were defined (see Hogquist Fig. 2) from which key conclusions are drawn.

To make the data more robust, 1) cell numbers must be included for all experiments;

1. rather than normalizing results to "the average H2-O WT levels", the actual data should be included;
2. figures should be more completely labeled/described;
3. FACS gating strategies should be clearly laid out (again, see Hogquist for examples). Furthermore, efforts must be made to explain why results are so different from analyses of H2-O deficient mice that have been published by many other groups. For example, the reported "dramatic increase in the proportion of CD3+CD4+ T cells" is not consistent with previous reports starting with Lars Karlsson's initial report (Liljedahl et al 1998). Extensive spontaneous activation of CD4 T cells has also not been reported in other papers that have studied these mice. Again, the paper is not placed in the context of the long, very thorough analysis of both the H2-O deficient mice and the study of H2-O/DO and H2-M/DM in general.

Reviewer #2 (Public Review):

Summary:

The manuscript's main claim is that the absence of H2-O, a component of the MHC II presentation pathway, promotes regulatory T cell development and function.

Unfortunately, the submitted material is not sufficient for proper evaluation of the manuscript, both in terms of the significance of the findings and the strength of the supporting evidence.

Major issues include:

- the scRNAseq (shown in Fig. 5) is too rudimentary to allow any conclusion. Statements in the text (eg "Principle Component Analysis (PCA) of the normalized scRNA-seq data identified 11 distinct CD4 T cell clusters", line 166) suggest that additional expertise should be leveraged for these analyses.

- Most flow cytometry data (Figs. 1 and 2) shows marginal (at best) differences on y-axis truncated bar graphs, with no original data plot, gating strategies, etc., severely challenging conclusions drawn from this data.

Author Response:

We are sorry that both eLife and the Reviewers feel that our submitted studies are currently insufficient to support our hypothesis that loss of H2-O function affects thymic Treg selection. As this is the first study directly evaluating loss of H2-O in the thymus we do not feel that we overstated our finding as suggested by Reviewer 1. We hope that a revised version of the manuscript can satisfy the reviewers' criticisms.

-Reviewer 1 is asking us to address the presumed discrepancies between our previous work (Welsh et al 2020, <https://doi.org/10.1371/journal.pbio.3000590>) and data from Lee et al. 2021 (<https://doi.org/10.4049/jimmunol.2100650>) in this current manuscript, which does not report on the development of EAE in DO-KO and DO-WT mice. All experiments here are on naïve mice. As such, we wish to justify our lack of discussion of Lee et al (2021) findings.

Lee et al (2021) reported the effects of DO on both EAE and SLE development, they used mainly H2-O β KO mice. As we have never used these CRISPR generated mice, we cannot have a direct in-house comparison. However, we did note that reported disease curve for female H2-O β KO mice had a similar trend indicating increased EAE disease development, similar to what we have reported back in our 2020 paper (Welsh et al PLoS Biology). In the single experiment that utilized H2-O β KO mice for EAE development, Lee et al found a different disease trend than ours. However, Lee et al (2021)'s tested only 4-5 mice per group in the single experiment and their measurement of the disease development solely relied on visual assessment of the limbs and tail functionality. Our study verified EAE disease development by multiple approaches including analyses of MOG-specific tetramer staining of the CNS CD4 lymphocyte infiltrate, and in vivo NIRF whole-body imaging on diseased DO-WT and DO-KO mice using an antibody probe specific to MBP. We had repeated our experiments on the disease development greater than 15 times using 5-8 mice per group. Below is an excerpt from our Results Section of Welsh et al PLoS Biology, clearly explaining how many experiments were performed and the number of mice per group per experiment:

"From these studies, we found that DO-KO mice had an accelerated onset of disease compared to DO-WT mice (Fig 7A). Disease symptoms (Score 1) appeared around Day 8–10 and quickly progressed to advanced disease (Score 3–4) by Day 14–16 in DO-KO. In contrast, DO-WT mice started showing symptoms around Day 12 and progressed to advanced disease scores by Day 20. Total cell infiltration into the CNS tissue was slightly higher in DO-KO mice, but no change in total brain weight was observed (S5 Fig). To further correlate the state of disease with CD4 infiltration, we performed in vivo NIRF whole-body imaging on diseased DO-WT and DO-KO mice using an antibody (Ab) probe specific to myelin basic protein (MBP). The Ab reacts with MBP only when the myelinated glia cells are damaged during disease development [56]. Thus, by detecting demyelination, whole-body imaging allowed us to fully visualize the co-localization of CD4 T cells at the sites of demyelination occurring in diseased mice. Interestingly, when mice of various disease scores were imaged, we found increased co-localization of infiltrating CD4 T cells with anti-MBP staining in DO-KO mice, but not in DO-WT mice (Fig 7B). These data not only confirmed the flow cytometric findings that diseased DO-KO mice have a greater influx of lymphocytes into their CNS tissue (S5 Fig), it also verified the massive demyelination that occurs during the disease"

And again in the Legend to Figure 7;

“Representative curves showing the time course of disease development in DO-KO (red) and DO-WT mice (white). N = 5 mice per group, representative of >15 repeat experiments. Score system: 0 = no symptoms, 1 = limp tail, 2 = limp tail + partial hind limb paralysis, 3 = limp tail + total hind limb paralysis, 4 = limp tail + total hind limb paralysis + partial forelimb paralysis. Data represented as mean $\pm$ SEM.”

Despite clarity of the description of our experiments, Lee et al have publicly slandered us and grossly misrepresented our work by stating the following:

“A recent study (11-Welsh et al) found that B6.Oa^{-/-} mice were more susceptible to EAE than control B6J animals. However, that conclusion was based on a single experiment, in which control B6J mice developed very mild EAE disease with an average score of 1, which is far lower than the disease scores published by other groups (30–32) and also observed in our study. Thus, in this inducible model of autoimmunity, H2-O deficiency does not contribute to either disease development or severity.”

-Another important variable between our studies and Lee et al (Lee et al 2021) was the use of a commercially available disease induction kit versus our immunization solutions that followed the established protocols by Nancy Ruddle et al (J Exp Med. 1997 Oct 20; 186(8): 1233–1240. doi: 10.1084/jem.186.8.1233). Notoriously, EAE disease development could vary widely based upon the quantities and purity of, a) MOG peptide, b) amount of tuberculosis antigen in the CFA, c) quantity of pertussis toxin and injection strategies, as well as many other uncontrollable factors. While a comparison these two results are irrelevant to our current study, we will be more than happy to compare our results from the previously published work with Lee et al. in the discussion.

-We want to emphasize that we did follow Hogquists et al’s gating strategy for detecting auditing vs deleted thymocytes by subdividing total thymocytes into “Non-signaled” (TCR- β -, CD5-/inter) and “Signaled” (TCR- β + CD5+/hi) populations before further gating on only medulla localized CD4 T cells. The “CCR7+ CD4+” label in Figure 1 was meant to orient the reader without overwhelming the figure with numerous flow plots. To address this concern, we will be including (1) updated Supplemental figures showing the complete gating strategy, (2) updated figure legends and text to emphasize the fact that auditing/deletion gating came from CD4 T cells which passed positive selection (i.e. TCR- β + CD5+/hi), and (3) including representative flow plots for all Figure 1 panels to the revise manuscript.

-Also, regarding “discrepancies between our data and Liljedahl et al 1998”;

H2-O KO mice used by Liljedahl et al were on a 129/Ola genomic background. The H2-O KO mice used for both of our papers have been completely backcrossed to C57BL/6J. Clearly, non-MHC genes contribute to the impacts of MHC proteins, yet how the 129/Ola genomic background could affect the H2-O genes remains to be discovered. And (B), no data was shown supporting their published statement below:

“The proportions of B cells as well as of CD4+ and CD8+ T cells in the lymph node, spleen, and thymus were similar in H2-Oa-deficient and wild-type mice (data not shown)”. (Liljedahl et al 1998).

Reviewer 2:

scRNA-Seq analysis was performed by the Computational Biology Computing Core at Johns Hopkins School of Medicine. We missed including this acknowledgement as our core facility does not request authorship or acknowledgements. The sentence has been edited for the correct terminology.

-About truncated bar graph, in the entire paper we have only two bar graphs, neither of which is truncated. So, we are puzzled by the reviewer's comment as to what figure he/she is referring to. -We would like to remind the Reviewer 2 that since DO works together with DM and functions differently on peptide of different sequences, the reported data on cumulative effects of DO in vivo have notoriously been rather minor. Especially, since our current study focuses on the naïve mice, major changes were not expected.

-Regarding leaving out gating strategies, we missed out on providing the gating strategies for all the figure in the original version. However, full FACS gating strategies have now been provided in the new supplemental figures and representative FACS plots have been added to ALL main figures.